

Tomiya Yamamoto

yamamoto.tomiya.yq4@ms.naist.jp



PERSONAL PROFILE

I am a PhD student in the Division of Materials Science, at the Nara Institute of Science and Technology. My research interests include the development of new materials using Machine Learning. Currently, I am conducting research on material development using generative models.

SKILLS

Digital and computer skills

- General skills: Machine learning, Deep learning, Generative Model (VAE, GAN, LLM, etc)
- Programming Languages: MATLAB, C#, C++, Python (Machine Learning : Pytorch, Tensorflow, Scikit-learn, Materials: Pymatgen, Xenonpy, PyXtal)
- Cloud-based Technologies: Git

Domain Expertise

- Basic knowledge in Inorganic Materials
- Computational Science Skills (Density Functional Theory (DFT) Calculations, VASP)

EDUCATION

04/2025 – Current	Doctoral Degree in Engineering <i>Nara Institute of Science and Technology (Nara, Japan)</i> the Division of Materials Science Data Science Course Materials Informatics Laboratory (PI: Prof. Mikiya Fujii)
03/2025 – Current	Research Internship Student(Non-Exchange) <i>Nanyang Technological University (Singapore)</i> School of Materials Science and Engineering (MSE) The Hip Laboratory (PI: Assoc. Prof. Kedar Hippalgaonkar)
10/2024 – 01/2025	Research Internship Student(Exchange) <i>Karlsruhe Institute of Technology (Baden-Wuerttemberg, Germany)</i> Institute of Theoretical Informatics (ITI) AiMat group (PI: T.T.-Prof. Pascal Friederich)
04/2023 – 03/2025	Master Degree in Engineering <i>Nara Institute of Science and Technology (Nara, Japan)</i> the Division of Materials Science Data Science Course Materials Informatics Laboratory (PI: Prof. Mikiya Fujii)
02/2023 – 03/2023	Research Student in Engineering <i>Nara Institute of Science and Technology (Nara, Japan)</i> the Division of Materials Science Materials Informatics Laboratory (PI: Prof. Mikiya Fujii)

- 12/2022 – 01/2023 *ILSC Melbourne (Victoria, Australia)*
English Course for 4 weeks
- 04/2022 – 09/2022 **Master Degree in Engineering**
Doshisha University (Kyoto, Japan)
Graduate School of Life and Medical Sciences
Major of Biomedical Engineering and Informatics
Human Informatics and Systems Laboratory (PI: Prof. Hiroyasu Tomoyuki)
I made the decision to shift my career focus because I realized that materials are a crucial element in the advancement of medical devices. I particularly thought that Materials Informatics would allow me to leverage my knowledge and experience in information technology at that time.
- 04/2013 – 03/2018 **Bachelor Degree in Engineering**
Doshisha University (Kyoto, Japan)
Faculty of Life and Medical Sciences
Department of Biomedical Sciences and Informatics
Human Informatics and Systems Laboratory (PI: Prof. Hiroyasu Tomoyuki)

RESEARCH EXPERIENCE

- 04/2025 – Current *Nara Institute of Science and Technology(Nara, Japan)*
Doctoral Thesis Research
- “Development of Generative Models for Photocatalyst Discovery”
under Prof. Mikiya Fujii
 - This research focuses on developing generative models to discover novel powder-type semiconductor photocatalyst materials with desirable properties
- 04/2025 – Current *Nanyang Technological University(Singapore)*
Internship Research
- “Development of Generative Models for Inorganic Material Crystal Structure”
under Assoc. Prof. Kedar Hippalgaonkar
 - This research focuses on developing generative models to propose novel crystal structures, taking into account symmetry and thermodynamic stability.
- 10/2024 – 01/2025 *Karlsruhe Institute of Technology (Baden-Wurttemberg, Germany)*
Internship Research
- “Large language models for the automatic extraction of data from scientific literature”
under T.T.-Prof. Pascal Friederich
 - This study aims use state-of-the-art large language models (LLMs) to extract tabular data from scientific literature.
 - In my work, I will deploy and fine-tune pretrained LLMs
- 02/2023 – 03/2025 *Nara Institute of Science and Technology (Nara, Japan)*
Master Thesis Research
- “Multi-Objective Generative Adversarial Networks for Inorganic Chemical Compositions”,
under Prof. Mikiya Fujii

- This study aims to propose novel inorganic chemical compositions that satisfy multiple desired properties by developing a multi-objective conditional Generative Adversarial Network
- To verify the accuracy of the generation process, I utilized Density Functional Theory (DFT) calculations and an ML model called CSPML, which can predict crystal structures from compositions, to obtain the crystal structures

Research Student Research

- “Drug Molecule Generation with Variational Auto Encoder”, under Prof. Mikiya Fujii
 - Generating effective novel drug molecules through gradient-based exploration of the latent space of Variational Auto Encoder

04/2021 – 09/2022

Doshisha University (Kyoto, Japan)

Bachelor Thesis Research under Prof. Hiroyasu Tomoyuki

- “The Effects of Auditory Stimulation on Episodic Memory: A Neuroscientific Analysis using fNIRS”
Under Prof. Hiroyasu Tomoyuki
 - Using fNIRS, which can measure blood flow in the brain, I investigated the relationship between music and long-term memory.

TEACHING AND WORK EXPERIENCE

07/2023 – 03/2025

MI-6 Ltd (Tokyo, Japan)

Job Internship

- Data analysis and machine learning work for chemical manufacturing industry

10/2022 – 11/2023

Nara Institute of Science and Technology

Teaching Assistant (TA)

- A demonstration and lecture on the characteristics and functions of molecules found in our surroundings (for junior high school students).

CONFERENCES(○: Presenter)

03/2023

The Chemical Society of Japan (Chiba, Japan)

Poster presentation

○**Yamamoto, T.**; Nishijo, M.; Harashima, Y.; Takasuka, S.; Takayama, T.; Sawada, Y.; Fujii, M.

Title: “Multi-Objective Generative Adversarial Networks for Inorganic Chemical Compositions”

05/2024

The 26th Annual Meeting of Japan Society of Theoretical Chemistry (Ibaragi, Japan)

Oral presentation

Nishijo, M.; Harashima, Y.; **Yamamoto, T.**; Takasuka, S.; Takayama, T.; Sawada, Y.; ○Fujii, M.

Title: “The Proposal of Novel Compositions Satisfying Desired Properties Using Generative Adversarial Networks”

09/2024

Symposium for Reaction Path Search 2024(Kyoto, Japan)

Poster presentation

○**Yamamoto, T.**; Harashima, Y.; Takasuka, S.; Takayama, T.; Sawada, Y.; Fujii, M.

Title: “Extrapolation Search for Inorganic Chemical Compositions Using Multi-Objective Conditional Generative Adversarial Networks”

- 09/2024 *The 18th Annual Meeting of Japan Society for Molecular Science*(Kyoto, Japan)
 Poster presentation
○Yamamoto, T.;Harashima, Y.; Takasuka, S.; Takayama, T.; Sawada, Y.; Fujii, M.
 Title: “Extrapolation Search for Inorganic Chemical Compositions Using
 Multi-Objective Conditional Generative Adversarial Networks”
- 09/2024 *The Japan Society of Applied Physics Student Capture Joint Research Conference*
 (Osaka, Japan)
 Poster presentation
○Yamamoto, T.;Harashima, Y.; Takasuka, S.; Takayama, T.; Sawada, Y.; Fujii, M.
 Title: “Extrapolation Search for Inorganic Chemical Compositions Using
 Multi-Objective Conditional Generative Adversarial Networks”

PAPERS

- T. Yamamoto, M. Nishijo, Y. Harashima, S. Takasuka, T. Takayama, Y. Sawada, M. Fujii, Multi Objective Generative Adversarial Network for Inorganic Chemical Compositions (Writing)
- M. Nishijo, T. Yamamoto, Y. Harashima, S. Takasuka, T. Takayama, Y. Sawada, M, Fujii Designing new materials by closed loop of GAN and crystal structure prediction (Writing)

AWARD

- 10/2024 **Best Poster Award**, The 18th Annual Meeting of Japan Society for Molecular Science(Japan)
- **JASSO Type 1 Scholarship Partial Repayment Exemption (Top 30%)**
 Granted by Japan Student Services Organization (JASSO) for outstanding academic performance during Master’s studies, 2023–2025
- **50% Tuition Fee Waiver, NAIST, 2025**
 Merit-based tuition reduction granted for outstanding academic performance

GRANTS and FELLOWSHIPS

- Scholarship for study program by the Nara Institute of Science and Technology, Japan
 (Living and travel expenses for a four-month research stay at KIT(Germany))
- Baden-Württemberg-STIPENDIUM scholarship 2024-2025, Germany
 (Living costs for a four-month research stay at KIT(Germany))
- Tobitate! (Leap for tomorrow) Study Abroad initiative STEAM course 16st batch by Ministry of Education, Japan
 (Living and travel expenses for a six-month research at NTU(Singapore))
- NAIST Granite Program Fellowship
 Nara Institute of Science and Technology, 2025–2028
 Monthly stipend (¥165,000) and annual research grant (¥400,000) for doctoral students demonstrating outstanding research ability (approx. S\$1,430/month and S\$ 3,480/year at ¥115/S\$)